

ProtoPNet Validation: Measuring Explainability and Performance

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A. Introduction

Explainable Artificial Intelligence (XAI) aims to make the decision-making processes of complex models more transparent and interpretable to humans. However, increasing model interpretability intrinsically can often come at the cost of predictive performance, creating a fundamental tradeoff between explainability and accuracy. However, post-hoc explanation methods such as saliency maps can often be misleading or uninformative.

This project explores the tradeoff in an intrinsic explanation method through the lens of ProtoPNet, a prototype-based neural network, that combines high classification performance with human-understandable visual explanations [1].

B. Data and model

We build upon an existing implementation of ProtoPNet and apply it to the CUB-200-2011 (Caltech-UCSD Birds) dataset, a fine-grained image classification benchmark commonly used in XAI research. Using this dataset, we develop and test a validation pipeline designed to quantify both model performance and the value of its explanations.

C. Preliminary analysis: cost of explanation

In a quick preliminary analysis in the project, we wanted to see if hacking out the prototype layer had an impact on the accuracy of the model. That is to say, we remove influence from the prototypes in the model completely.

When we trained and evaluated both types of networks in equal environments, an absolute increase in accuracy of around 3.5 percentage points was observed. This gives rise to the fact that the prototypes do have an impact on the model. We wish to explore methods based around the prototype layers and see how it can help explain the model.

D. Validation method: the framework

Following is our take on an abstract definition of an explanation. Given a model $F : \mathcal{X} \rightarrow \mathcal{Y}$ and an input $x \in \mathcal{X}$, an explanation $A(x; F)$ is a lower-dimensional, interpretable projection of the causal computational pathway that led to the prediction $F(x)$

$$A(x; F) = P \circ T(x; F)$$

where $T(x; F)$ encodes the mathematical trace of computations contributing to $F(x)$, and P is a projection onto an interpretable subspace $\mathcal{Z} \subset \mathbb{R}^k$ with $k \ll \dim(T)$.

E. Assessing explanation value

The non-trivial part of deriving a "good" explanation lies in the dimensionality reducing projection P which gives rise to the following pitfalls: (1) The projection P may distort or hide important aspects of the underlying high-dimensional trace T , thereby introducing bias or misrepresentation into the resulting explanation. (2) Even if the projection P effectively reduces dimensionality, it may still fail to align with human interpretability, rendering the explanation uninformative and, ultimately, useless.

Thus, a "good" explanation is one that exhibits **both** (1) high fidelity (i.e., minimal information loss by the projection) and (2) high human interpretability, where a lack of either will not produce a useful explanation.

The following measures aims to assess the above stated criteria for a good explanation and while not being able to measure every aspect of the criteria in full, arguably correlates sufficiently well with the performance on the criteria themselves.

Measuring fidelity We assume that the convolutional layers of the ProtoPNet aren't affected by the models goal of achieving good explainability which is a crude assumption given ProtoPNet intrinsically embedding the explainability in the model via the loss function and common backpropagation. For a given test sample x and a model prediction $\hat{y}$ we attempt to reconstruct the activations of the layer preceding the prototype layers (denote this layer $\ell_{(-1)}$) solely based on the prototype activations. We measure the reconstruction error to the actual activations of $\ell_{(-1)}$ by the MSE across layer nodes and channels.

Measuring human interpretability While there are multiple aspects that factor into making an explanation interpretable to humans we chose to focus on a single yet important factor: to make an explanation interpretable it must consist of just a few "concepts", which for the case of ProtoPNet means that a classification should rely on just

a few prototypes and not many. For each correctly predicted test sample, we determine which prototypes contribute positively to the predicted class by computing an evidence score defined by the product of the prototype activation and its final layer weight. We then measure what percentage of the total positive evidence top-k prototypes. High concentration percentages indicate that the model’s decision relies on only a few key prototypes, making the explanation simpler and more interpretable to humans. We only consider correctly predicted samples to isolate the performance of the explainability layer of the model from the overall classifier.

F. Experimental setup and results

Fidelity For testing fidelity we define a simple Fully Connected Neural Network model and task it with learning the mapping from the prototype activations to the preceding non-prototype layer $\ell_{(-1)}$:

$$R : \mathbb{R}^m \rightarrow \mathbb{R}^{d \times h \times w}$$

where m is the number of prototypes, and $C \times H \times W$ are the dimensions (channels, height, width) of the activations in layer $\ell_{(-1)}$, $\ell_{(-1)}$ being a convolutional layer. The reconstruction error is then measured as:

$$\mathcal{L}_{\text{Fidelity}}(A) = \frac{1}{H \cdot W \cdot C} \sum_{h=1}^H \sum_{w=1}^W \sum_{c=1}^C \left(\mathbf{Z}_{h,w,c} - \hat{\mathbf{Z}}_{h,w,c} \right)^2$$

where $\mathbf{Z}$ represents the true activations of layer $\ell_{(-1)}$ for a given input, $\hat{\mathbf{Z}}$ represents the reconstructed activations from the prototype activations. For our model $H = W = 7$, $C = 128$.

The reconstruction model R is implemented with single hidden layer followed by ReLU activation. We train the model on the training set to avoid memorization, and train it for 10 epochs with a batch size of 50 and the Adam optimizer.

The model converges to a MSE of **0.022364**.

Interpretability For each correctly predicted test sample x with predicted class $\hat{y}$, we compute the evidence vector $\mathbf{e} \in \mathbb{R}^m$ where:

$$e_i = s_i \cdot w_{\hat{y},i}$$

with s_i being the similarity of prototype i and $w_{\hat{y},i}$ the corresponding final layer weight. We measure evidence concentration as:

$$\rho_k = \frac{\sum_{i=1}^k e_{(i)}}{\sum_{j:e_j > 0} e_j}$$

where $e_{(i)}$ is the i -th largest evidence value and $k \in \{1, 3, 5\}$.

We find that for correctly predicted samples, on average **31.51%** of evidence is in the top 1, **66.70%** of evidence is in the top 3 and **85.21%** of evidence is in the top 5.

G. Discussion: strengths and weaknesses of validation methods

For fidelity, the relatively low MSE of around 0.022 indicates that the prototype activations capture a substantial portion of the information present in the preceding convolutional layer. This suggests that the dimensionality reduction from the full feature space to the prototype space introduces just minimal information loss. In other words, the prototype layer preserves most of the discriminative features necessary for classification, demonstrating high fidelity. However, it’s important to note that this assumes the convolutional layers themselves are unaffected by the explainability objective which may not fully hold given ProtoPNet’s joint optimization of both classification and interpretability through its loss function.

For the human interpretation, the results demonstrate strong evidence concentration among the top prototypes, indicating that ProtoPNet produces relatively sparse and interpretable explanations. The fact that approximately two-thirds of the decision evidence comes from just 3 prototypes suggests that the model’s reasoning can be effectively communicated to humans through a small set of concrete visual examples. We can see a linear increment in the percentage of evidence in the top prototypes for correct predictions, meaning, the more top prototypes are used, the more evidence that comes from them. This directly suggests the importance of the prototypes.

H. Future work

For future validations, we can attempt to validate the Robustness. This idea is motivated by paper [2] and its definition of robustness. We would remove the top 5 prototypes, then 10, that the model says are important, in order to see if the accuracy drops. If the explanations are faithful, they should drop, if they don’t, then the explanations are erroneous and the prototypes simply appeared decent.

Annexes

For additional materials and code, see the project repository.

References

- [1] Chaofan Chen, Oscar Li, Chaofan Tao, Alina Jade Barnett, Jonathan Su, and Cynthia Rudin. This looks like that: Deep learning for interpretable image recognition, 2019. [1](#)
- [2] Zixi Chen, Varshini Subhash, Marton Havasi, Weiwei Pan, and Finale Doshi-Velez. What makes a good explanation?: A harmonized view of properties of explanations, 2024. [2](#)